

DATABASE CONCEPTS: UUID, ENUM, AND SET

What is UUID?

UUID stands for

Universally Unique Identifier

It is a 128-bit value designed so that the probability of two independently generated UUIDs being identical is extremely low.

Unlike

```
1  
2  
3  
4  
5
```

UUID looks like

```
550e8400-e29b-41d4-a716-446655440000
```

Why Do We Need UUID?

Suppose Amazon has servers in:

- Mumbai
- Delhi
- Bengaluru
- Chennai

Each server creates orders independently.
If every server uses

AUTO_INCREMENT

then

Mumbai

1
2
3

Delhi

1
2
3

Both generate Order ID 1.
Collision.

With UUID

Mumbai

550e8400-e29b-41d4-
a716-446655440000

Delhi

9d4a2b60-80b3-4a73-
a5d1-52d4d63ef001

No collision.

UUID Structure

Example

550e8400-e29b-41d4-a716-446655440000

Broken into groups

8-4-4-4-12total

- 32 hexadecimal digits

- or
- **128 bits**

Storage

There are two common ways.

As CHAR

```
UserID CHAR(36)
```

Stores

```
550e8400-e29b-41d4-a716-446655440000
```

Storage

36 characters

including hyphens.

As BINARY

```
UserID BINARY(16)
```

Stores the raw 128-bit value.

Storage

16 bytes

Much smaller.

Much faster.

Many production systems store UUIDs in binary form and convert them to text only when needed.

Insert Example

```
CREATE TABLE Users(  
    UserID CHAR(36),  
    Name VARCHAR(100)  
);
```

```
INSERT INTO UsersVALUES  
(UUID(), 'Aadya');
```

UUID Versions

There are multiple versions.

Version 1

Based on

- timestamp
- machine identifier
- clock sequence

Advantages: Time ordered.

Disadvantages: Can reveal machine/time information.

Version 3

Generated using

- namespace
- MD5 hash

Deterministic.

Version 4

Most common.

Generated using random numbers.

Example

```
9f4fd9d2-6f4e-4d49-b71d-8c34f4c7d2ab
```

Almost every backend project uses Version 4 unless ordering is important.

Version 5

Uses SHA-1 hashing.

Deterministic.

UUID Indexing Impact

UUID

```
9f....
```

```
2b....
```

```
8d....
```

```
a1....
```

Random positions.

Database constantly inserts rows into different parts of the index.

This causes:

- page splits,
- index fragmentation,
- increased I/O.

This is one of the biggest disadvantages of random UUIDs (especially Version 4).

Security Benefit

If a hacker sees

```
/users/5
```

they can guess

```
/users/6
/users/7
/users/8
```

This is called ID enumeration.

UUID

```
/users/550e8400-e29b-41d4-a716-446655440000
```

Impossible to guess by simple incrementing.

Much better for public-facing APIs.

What is ENUM?

ENUM stands for Enumeration.

It allows a column to store only one value from a predefined list.

Think of it as a drop-down menu

Syntax

```
CREATE TABLE Student(
    StudentID INT PRIMARY KEY,
    Name VARCHAR(100),
    Gender ENUM('Male', 'Female', 'Other')
);
```

Insert

```
INSERT INTO StudentVALUES
(1, 'Aadya', 'Female');
```

Gender ENUM('Male','Female','Other')

Internally

MySQL stores

Male	→ 1
Female	→ 2

Other	→ 3
-------	-----

The database stores the numeric index,
not the full text,
while displaying the corresponding string.

Only **1 byte** is needed.

Very efficient.

Number of ENUM Values

Number of Values	Storage
Up to 255	1 byte
256–65,535	2 bytes

What is SET?

SET is another MySQL-specific datatype.

Unlike ENUM,

a row can contain multiple values from the predefined list.

Example

```
CREATE TABLE Employee(
  Skills SET(
    'C++',
    'Python',
    'SQL',
    'Java'
  )
);
```

```
INSERT INTO EmployeeVALUES
( 'C++', SQL '');
```

Internal Representation of SET

Suppose

- C++
- Python
- SQL
- Java

Internally

C++	0001
Python	0010
SQL	0100
Java	1000

Advantages and Disadvantages

Advantages of ENUM

- Prevents invalid values.
- Compact storage.
- Easy to understand.
- Fast comparisons.
- Simple schema for fixed choices.

Advantages of SET

- Stores multiple selections.
- Efficient bitmask storage.
- Convenient for a small, fixed set of options.

Disadvantages of ENUM

- Adding values requires a schema change.
- MySQL-specific (less portable to other databases).

Disadvantages of SET

- MySQL-specific.
- Harder to query and maintain than a normalized many-to-many relationship.